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🧠 MCP Quiz (Model Context Protocol)

🔹 Section A: MCQs (1 Mark Each)

1. What is MCP primarily used for?
A. Data visualization
B. Agent-to-tool communication
C. Database management
D. UI design

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2. MCP stands for:
A. Model Control Pipeline
B. Machine Context Processing
C. Model Context Protocol
D. Managed Compute Platform

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3. In MCP architecture, what is the role of an "Agent"?
A. Stores data
B. Executes database queries
C. Decides actions and interacts with tools
D. Designs UI

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4. Which component actually performs tasks like API calls?
A. Agent
B. Tool

C. Context
D. Connector

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5. MCP helps in:
A. Replacing APIs
B. Standardizing communication between AI and tools
C. Removing cloud dependency
D. Building databases

🔹 Section B: True / False (1 Mark Each)

6. MCP eliminates the need for tools.

FALSE

7. Agents can call multiple tools using MCP.

TRUE

8. MCP only works with Python.

FALSE

9. Context in MCP helps the agent make better decisions.

TRUE

10. MCP architecture includes Agent, Tool, and Context.

TRUE

🔹 Section C: Short Answer (2 Marks Each)

11. Define Model Context Protocol (MCP).

A standardized framework that enables AI agents to communicate with external tools and systems efficiently using structured context.

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12. What is “context” in MCP?

Context is the structured information (user input, history, environment, memory) that helps the agent understand and make decisions.

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13. Give two examples of MCP tools.

Two examples of MCP tools:

- Weather API
- Database query tool

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14. What is the difference between an Agent and a Tool in MCP?

Difference between Agent and Tool:

- Agent: Decides what to do and when
- Tool: Executes specific tasks (e.g., API calls, calculations)

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15. Why are connectors important in MCP?

The connectors are important because connectors link agents to tools and ensure standardized, secure, and compatible communication.

🔹 Section D: Scenario-Based Questions (3 Marks Each)

16. Scenario:
An AI assistant receives a user query: *“What is the weather in Delhi?”*

- Identify the role of the agent ->

  Understands the query and decides to fetch weather data

- Identify the tool used ->
  Weather API tool

- Explain how MCP enables this interaction->

  Provides a structured way for the agent to call the tool and receive results

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17. Scenario:
A chatbot needs to fetch customer data from a database and then calculate a discount.

- How many tools are involved?

  Two tools : database tool and calculation tool
- What is the role of MCP here?

  Role of MCP is to Coordinate communication between agent and both tools, ensuring smooth data flow

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18. Scenario:
An agent fails to provide accurate output due to missing context.

- What went wrong?

  Agent lacked sufficient context (e.g., user data/history)
- How can MCP fix this issue?

  Provide richer structured context so the agent can make accurate decisions

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🔹 Section E: Advanced / Conceptual (5 Marks Each)

19. Explain MCP architecture with its key components.

MCP Architecture:
MCP consists of four main components:

- Agent: The decision-maker that interprets user input and selects actions

- Tool: External systems/APIs that perform tasks

- Context: Information that guides the agent's decisions
- Connectors: Bridge between agent and tools

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20. Describe how Agent–Tool communication works in MCP step by step.

Agent–Tool Communication (Step-by-Step):

- User sends a query
- Agent interprets the query
- Agent checks context
- Agent selects the appropriate tool
- MCP formats the request
- Tool executes the task
- Tool returns the result
- MCP standardizes the response
- Agent processes and delivers the final answer